

Analysis First-Year Exam, January 2026, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Prove that the bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable in each of the following cases:
 - (a) f is increasing;
 - (b) f is continuous.
2. Let $(f_n)_{n=0}^\infty$ be a uniformly convergent sequence of continuous real-valued functions on the metric space X and let $(x_n)_{n=0}^\infty$ be a convergent sequence in X . Must the real sequence $(f_n(x_n))_{n=0}^\infty$ converge? Proof or counterexample.
3. Let $\mathcal{F}$ be an equicontinuous family of real-valued functions on the connected metric space X and when $x \in X$ write $\mathcal{F}_x = \{f(x) : f \in \mathcal{F}\} \subseteq \mathbb{R}$. Prove: if $\mathcal{F}_x$ is bounded for some $x \in X$ then $\mathcal{F}_x$ is bounded for every $x \in X$.
4. Let $X \subseteq \mathbb{R}$ have finite Lebesgue measure $\lambda(X)$. Prove:
 - (a) for each $\varepsilon > 0$ some open set $U \subseteq \mathbb{R}$ satisfies $U \supseteq X$ and $\lambda(U) < \lambda(X) + \varepsilon$;
 - (b) some Borel set $B \subseteq \mathbb{R}$ satisfies $B \supseteq X$ and $\lambda(B) = \lambda(X)$.
5. Let the sequence $(f_n)_{n=0}^\infty$ of non-negative integrable functions converge pointwise to f on some measure space $(\Omega, \mathcal{A}, \mu)$. If it is assumed that the integrals $\int_\Omega f_n d\mu$ are all at most $M < \infty$, does it follow that f is integrable and that $\int_\Omega f d\mu \leq M$? Proof or counterexample.